ASAD HUSSAIN



Embedded Systems | Digital Design (Verilog) | PCB Design | IoT | Applied AI/ML

Islamabad, Pakistan • +92 320 4141092 • asadh1521@gmail.com
Portfolio | GitHub

PROFESSIONAL SUMMARY

Electrical Engineering student at NUST (B.E., expected 2027; CGPA 3.21/4.00) specializing in **embedded firmware, PCB design and Verilog-based digital design**. Delivered an ESP32 fleet-management platform (GPS, RFID, MQTT, Flask/SQL Server) at Heavy Industries Taxila (HIT), laid out boards in Altium Designer and EasyEDA, and designed and bench-validated analog circuits. Adds applied AI/ML and backend experience from remote internships; current research covers EV battery monitoring and range estimation.

EDUCATION

National University of Sciences and Technology (NUST) — B.E. Electrical Engineering *2023 – 2027 (expected)*

CGPA: 3.21 / 4.00 • Coursework: Digital System Design, Embedded Systems, Digital Communications, DSP, Analog Electronics, Power Electronics, Control Systems

TECHNICAL SKILLS

Embedded & IoT	C/C++, ESP32, GPS/RFID interfacing, MQTT, Wi-Fi/Bluetooth, GSM telemetry, sensor interfacing (MAX30102, MPU6050, MLX90614)
PCB & Electronics	Altium Designer, EasyEDA, Proteus, schematic capture and board layout, analog design, simulation and bench testing
Digital Design & DSP	Verilog, MATLAB, digital logic, DSP, communication-system modeling, forward error correction
Backend & Tools	Python, Flask, FastAPI, SQL/SQL Server, REST APIs, Docker, Git
AI / ML	Scikit-learn, CNNs, classification, LangChain, FAISS, retrieval-augmented generation, YOLOv8
Battery Systems	Current/voltage sensing, SOC/SOH estimation, cell protection and balancing, cloud telemetry

EXPERIENCE

Automation Engineer - Cymatics Technologies (Full-time) *Sep 2026 – Present*

- Built and deployed Bid Monitor on Render using Python/FastAPI, Playwright collectors, HTML/PDF extraction and a searchable dashboard.
- Integrated Google Sheets storage, record deduplication, GitHub Actions scheduling and reviewed n8n outreach.

Research Assistant, EV Battery Management Systems — NUST *Aug 2026 – Present*

Supervisor: Dr. Hassan Khalid

- Designed a battery-monitoring circuit combining current and voltage sensing with an AI model that predicts energy consumption and remaining driving range.
- Researching cell monitoring, protection, balancing, SOC/SOH estimation and cloud telemetry for remote battery analytics.

Machine Learning & Backend AI Engineering Intern — FlyRank AI (Remote) Jul - Sep 2026

- Built a content-opportunity scoring pipeline that flags pages with strong search visibility but low click-through rates and turns model output into prioritized content-refresh recommendations.
- Developed an authentication-protected, database-backed CRUD API, containerized it with Docker and automated the scraping workflow feeding it.
- Completed an accepted capstone project on LLM usage metering and billing.

Technical Intern — NEPRA, Islamabad *Jul 2026 – Aug 2026*

- Analyzed NGC data for **233 grid-station transformers** and found **116 (about 50%)** operating above 80% of rated capacity, highlighting heavily loaded assets.
- Mapped generation, transmission, distribution and supply licensing processes; studied smart-metering communications (DLMS/COSEM) and transformer protection.

Machine Learning Intern — CodeAlpha (Remote) *Jun 2026 – Jul 2026*

- Delivered **three end-to-end ML projects** (credit scoring, handwritten-character recognition, disease prediction), evaluating each classifier with precision, recall, F1 and ROC-AUC.

Engineering Intern, Embedded Systems — RDC, Heavy Industries Taxila (HIT) *Jun 2025 – Aug 2025*

- Delivered an **ESP32-based fleet-management platform** integrating GPS tracking, RFID driver authentication, cellular MQTT telemetry and a Flask/SQL Server backend.
- Wrote C/C++ state machines for **1 Hz GPS capture** and driver feedback; implemented geofencing, harsh-driving detection and SD-card data resilience.
- Built a Leaflet.js/Chart.js dashboard to visualize vehicle tracking and driving-behavior data.

Research Intern — Robot Design & Development Lab, NCRA, NUST *Jul 2024 – Aug 2024*

- Implemented ESP32 Wi-Fi and Bluetooth communication, wireless command handling and device control during a six-week embedded-systems and IoT internship.

PROJECTS

Lite Semantic Communication - Final Year Project (in progress) *In progress*

- Developed a semantic-communication model with forward-error-correction encoding/decoding in MATLAB to study message reconstruction under channel noise; DSP-kit integration planned next.

4-Layer Custom PCB Design *4-layer PCB*

- Designed a custom **4-layer PCB** end to end: schematic capture, layer stack-up planning, component placement, routing and fabrication output generation.

ESP32 Interface and Sensor Boards — Altium Designer, EasyEDA *PCB Design*

- Designed an ESP32 USB communication interface board in Altium Designer and a two-board sensor/GSM telemetry system in EasyEDA.
- Integrated MAX30102, MPU6050 and MLX90614 sensors into the sensor-board design, pairing measurement acquisition with GSM telemetry.

Multi-Stage Audio Power Amplifier — Analog Electronics *Simulated & built*

- Designed, simulated and **implemented in hardware** a three-stage amplifier (common-source preamplifier, common-emitter gain stage, diode-biased Class AB output) with **694 V/V** overall gain across 20 Hz – 20 kHz.

AMI-Based IoT Smart Energy Meter — ESP32, Proteus, Web Dashboard *Prototype*

- Built an energy-metering prototype linking an ESP32/LilyGO sensing node to a web dashboard for electricity-consumption readings; contributed sensor interfacing and dashboard integration.

Analog Filter and Regulated DC Supply — Circuit Design and Bench Testing *Bench-validated*

- Simulated, built and validated a second-order Butterworth low-pass filter (flat 20 dB passband to 900 Hz, **>10 dB attenuation at 11.25 kHz**) and a regulated DC power supply.

Retrieval-Augmented Knowledge Bot — atomCamp AI Bootcamp *AI/ML*

- Built and deployed a knowledge-bot agent using LangChain, FAISS and OpenAI APIs with a Streamlit interface, packaged and served with Docker and FastAPI.

Additional academic work: 4-bit ALU (Verilog), EEG-cleaning MATLAB GUI, five-band graphic equalizer, PID DC-motor controller, TDA7000 FM receiver.

KEY STRENGTHS

- **End-to-end hardware–software integration:** PCB layout and ESP32 firmware through MQTT/cloud backends and web dashboards (HIT fleet platform, IoT smart energy meter).
- **Hands-on builder:** designs, simulates, builds and bench-tests circuits, from analog amplifiers and filters to regulated supplies and 4-layer boards.
- **Data-driven engineering:** applies analysis and ML to real systems, such as grid-transformer loading at NEPRA (233 assets), battery range prediction and content scoring.
- **Leadership and teaching:** Technical Lead of the NUST Robotics Society; organizes PCB and embedded workshops and demonstrates projects at public events.
- **Fast, self-driven learner:** five professional courses and certifications (Altium, IBM, MathWorks, Deloitte, atomCamp) alongside full-time coursework; NMUN delegate with strong written and spoken English.

TRAINING & CERTIFICATIONS

Atomcamp AI Bootcamp | Jun-Nov 2026 (in progress)
Docker/FastAPI, n8n

- **IBM** — Data Analysis with Python (Apr 2024)
- **MathWorks** — MATLAB Fundamentals
- **Altium Education** — PCB Basic Design (2026)
- **Deloitte WorldClass** — Effective Leadership (Aug 2026, 90%)

LEADERSHIP & ACTIVITIES

- **Technical Lead, NUST Robotics Society** — led a live YOLOv8 object-detection dashboard demonstration at Voltfest 2026, using an ESP32 with a webcam as the video source, and organized PCB and embedded-systems workshops.
- SEMPEC project-display participant (PID motor controller) • IEEE member • NMUN 2024 delegate

Languages: Urdu (native) • English (professional working proficiency) • French (elementary)